(Please write your Enrollment Number)

End-Term Examination
(B. Tech CSE AI-CBCS)(SUBJECTIVE TYPE)
(April, 2022)

Subject Code: BAI-103	Subject: Computer Organization And Architecture	
Time : 3 Hours		Maximum Marks : 60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.		

Q1.		(2.5*8=20)
	(a) Simplify the boolean expression : $y = \left[\left(\bar{A}+\bar{B}\right)(A+B)\right]$	
	(b) Convert 461 (decimal) to its equivalent octal number.	
	(c) What do you mean by overflow?	
	(d) Explain the basic format of floating point number with an example.	
	(e) Write steps in fetching a word from memory.	
	(f) What is CISC & its characteristics?	
	(g) What is an interrupt?	
	(h) What are the modes of data transfer?	
	UNIT-1	
Q2.	(a) Determine by the means of truth table the validity of De-morgan's theorem for the three variables. **(4marks)** $(ABC)' = A'+B'+C'$ (b) Simplify the following expression using Boolean Algebra. **(2marks)** $AB+A(CD+DC')$ (c) Draw the logic Diagram and truth table of a 2 to 4 decoder with NAND Gate. **(4marks)**	**(10marks)**
Q3.	(a) Describe decoder Expansion and draw 3 x 8 decoder from 2 x 4 decoder. **(8marks)** (b) Draw the block diagram of 4 to 1 line multiplexer and explain its operation by means of a function table. **(2marks)**	**(10marks)**
	UNIT-2	
Q4.	Describe the types of instructions on the basis of address fields used in the instruction with example.	**(10marks)**
Q5.	Evaluate the arithmetic statement X = (A+B) * (C+D) using a general register computer with three address, two address, one address instruction format to evaluate the expression.	**(10marks)**
	UNIT-3	
Q6.	(a) What is parallel processing? **(2marks)** (b) What is Array Processing & explain the two types of array processors. **(5marks)** (c) Draw the hardware for signed- magnitude addition and subtraction. **(3marks)**	**(10marks)**
Q7.	(a) Draw the hardware for multiply operation for signed-magnitude data. **(3marks)** (b) Show with successive shift method, the multiplication of two- fixed point binary numbers: 10110 → multiplicand 10010 → multiplier. **(3marks)** (c) Draw the flow chart for multiply operation. **(4marks)**	**(10marks)**
	UNIT-4	
Q8.	(a) Explain Handshaking with block diagram. **(4marks)** (b) Explain the working of DMA controller with the block diagram. **(6marks)**	**(10marks)**
Q9.	(a) What is an associative memory? **(2marks)** (b) Give a brief summary of the peripheral devices. **(4marks)** (c) Explain with the help of the diagram the connection of I/O bus to input-output devices. **(4marks)**	(10marks)